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Academic Year: 2021-2022

Programme: B. Tech Computer Science & Business Systems Year: I Semester: II

Subject: Data Structures & Algorithms

Date: 16 May 2022

Marks: 100
Time: 11.00 am to 2.00 pm
Durations: 3 (Hrs)
No. of Pages: 03

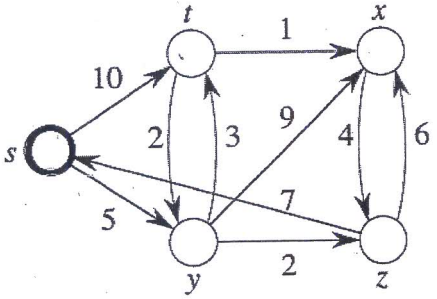
Final Examination (2021-22) / Re-Examination (2020-21)

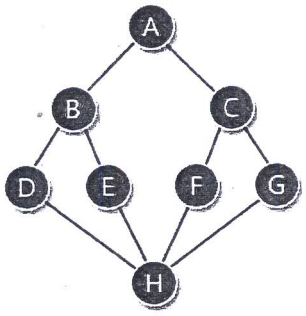
Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. 1 is compulsory.
2) Out of remaining questions, attempt any 4 questions.
3) **In all 5 questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

Q1
CO-1;
BL-3
	a.	Explain Big O notation. Compute the time complexity of the following code
		int fun (int n)
		{
		int a[100], int sum = 0;
		for(int i =1; i< n; i++)
		{
		sum = sum + a[i];
		}
		return sum;
		}
			[5]

CO- 2; BL- 2	b.	State two applications of linked list. Compare linked list with an array.	[5]
CO- 3; BL-4	c.	Construct the expression tree for the following expression A + B * (C / D + E) + F	[5]
CO- 4; BL-3	d.	Write the algorithm for performing binary search on the sorted array.	[5]
Q2 CO-3; BL-4	a	The post order traversal of a binary tree is 9, 10, 7, 8, 5, 6, 3, 4, 2. The inorder traversal of the same tree is 9, 7, 10, 5, 8, 3, 6, 2, 4. i. Construct the binary tree from traversals. Show all steps ii. Find the height of the above binary tree. iii. Find the preorder traversal of the tree. iv. Find the number of leaf nodes in a tree	[10]

CO-2 BL-3	b	The software is made to handle customer request one by one of restaurant's take away service. Write an algorithm to implement insertion and deletion function of appropriate data structure which their team might have used in the software.	
Q3 CO-3 BL- 3	a	Following graph represent machines on the network. Apply Dijkstra's algorithm to find the optimal path between source machine ***s*** to all other machines to send data.  	[10]
CO-3; BL-2	b	Define recursion. Discuss the concept of merge sort with an example	[10]
Q4 CO-2; BL-3	a	You are hired to develop an application that stores patient details such as patient id and name in singly linked list. The application must perform below operations: i. Insert a new patient at the starting of the list. ii. Exchange nodes of first patient and last patient Write functions to implement above functionality.	[10]
CO-4; BL-3	b	Consider a hash table with size = 10. Using linear probing insert the following keys into the table. Demonstrate the working by showing each step. 28, 73, 64, 43, 37,19, 57, 28 and 102	[10]
Q5 CO-2; BL-4	a	Convert the following infix expression to postfix expression using stack. Show all stack steps. Evaluate the postfix expression using P =12 Q =3 R =2 S =2 T =1 U =2 ((P - (Q + R) * S) ^ (T + U))	[10]
CO-4; BL-3	b	A certain sorting technique was applied to the following data set 47, 3, 65, 38, 56, 92, 17, 9, 11 After two passes, the rearrangement of the data set is given as below 3, 47, 65, 38, 56, 92, 17, 9, 11 i. Identify the sorting algorithm that was applied. ii. Complete the remaining passes. iii. Write a function to implement the identified sorting algorithm	[10]

Q6 CO-2 BL-3	a	Perform Breadth first search traversal and write an algorithm for the same 	[10]
CO-4; BL-2	b	Write short note on a) file organization b) threaded binary tree	[10]